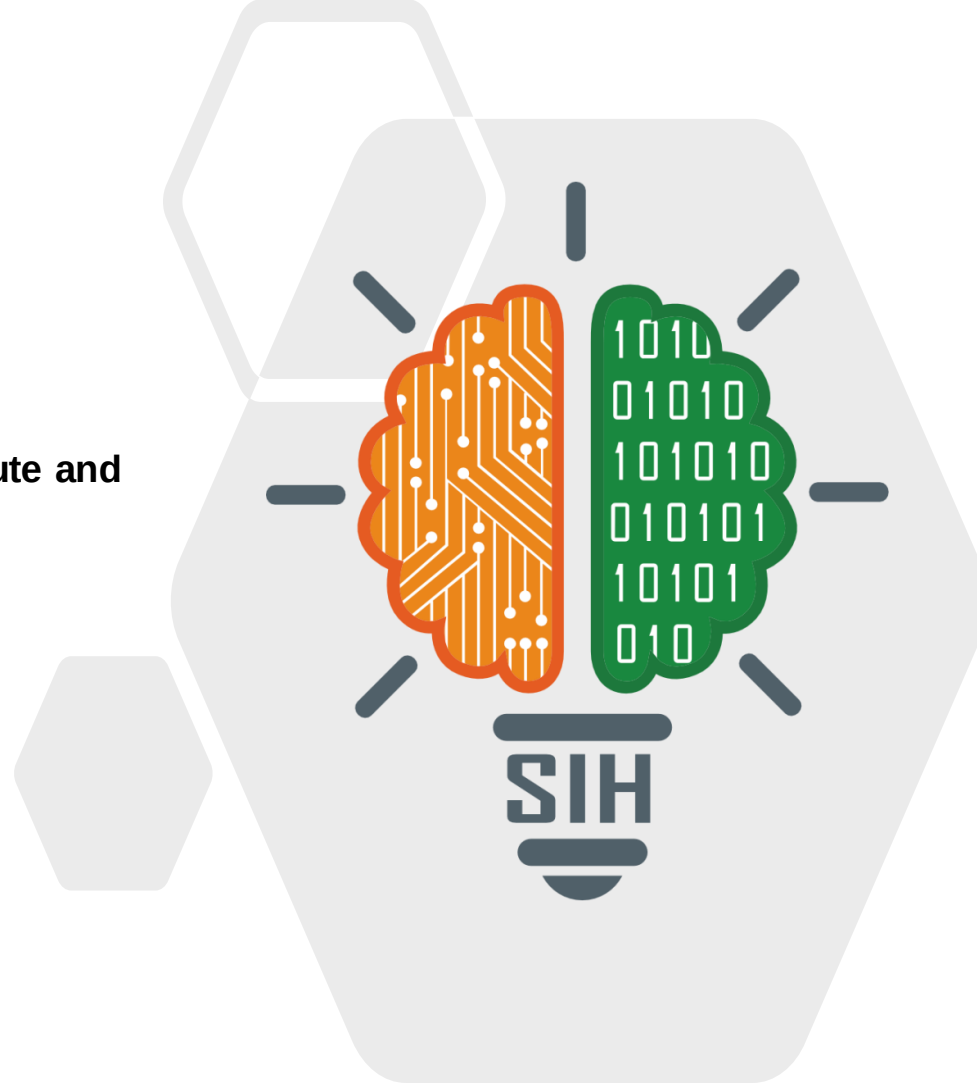


SMART INDIA HACKATHON 2024



- Problem Statement ID – SIH1610
- Problem Statement Title – Learning App for Deaf And Mute and sign language-English/Gujarati converter
- Theme - Smart education
- PS Category - Software
- Team ID - 37751
- Team Name (Registered on portal) - Leviosa



IDEA TITLE

Challenges

- The NEP mandates **inclusive education** with tailored tools for **63 million** individuals in India with hearing loss, focusing on ISL and local languages.
- **Parents** lack knowledge of ISL.
- **Teachers** struggle to offer ongoing support.
- **Employers** face challenges in integrating deaf individuals, leading to missed opportunities.

Value proposition

- Engaging **content in Gujarati Sign Language** for better comprehension.
- Tools for parents to support their **children's education**, regardless of ISL knowledge.
- **Integrated LMS** for task assignments and progress tracking.
- Essential language and communication skills for **improved employability**.

Proposed Solution

- Creating a Real-Time Sign Language Learning and Translation App to address educational challenges. It includes a Learning Management System (**LMS**) for assignments and progress tracking, aiding parents, teachers, and HR professionals in **communication** and resource access.

Key feature

- Video Pre-Processing with Sign Language Translation
- Online Meeting Support

Uniqueness

- Interactive **AI/ML Chatbot**
- Translating YouTube videos and integrating them with sign language.

Programming Languages:

Python: For developing and training ML models.

Dart: To build the app using Flutter.

Frameworks and Libraries:

TensorFlow: To train and deploy ML models.

Flutter: Cross-platform for mobile app development.

Firebase: Backend services, including authentication, database and storage.

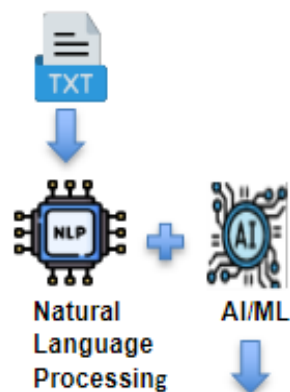
Blender: 3D animation tool

[\(View prototype\)](#)

The product is **50% complete**, and we are currently focused on developing further features of the app.

Key Features of the Leviosa App

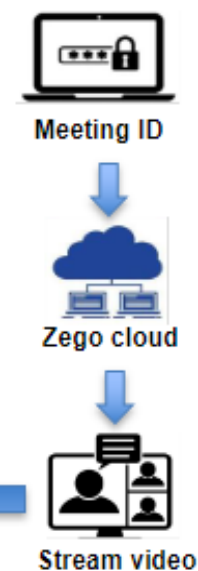
Text - Sign



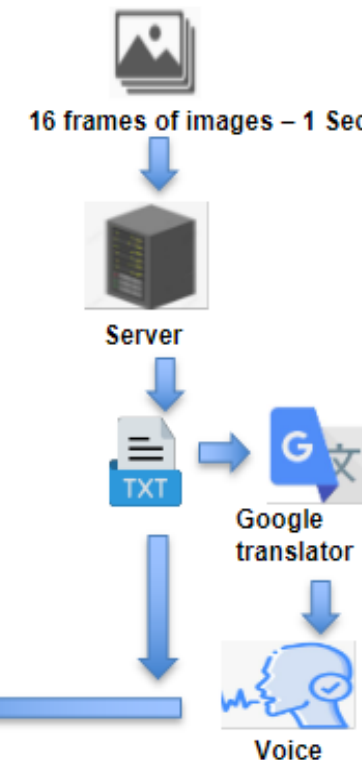
AI chat box



Meeting



Sign - Text



Analysis of the feasibility of the idea

- Existing frameworks support cross-platform development.
- Scalable across schools in Gujarat, with future expansion potential.
- User-friendly interface for students, teachers and parents.
- Promotes inclusivity and benefits specially-abled students.

Challenge 1

Limited Access to Data:

Difficulty in sourcing **accurate and comprehensive** Indian Sign Language datasets, especially for regional languages like Gujarati.



Solution

Data Access:

Partner with institutions or NGOs specializing in ISL to source and develop accurate datasets. Use **crowdsourcing techniques** to expand regional language datasets.

Challenge 2

Technology Limitations:

Challenges with speech-to-sign language **conversion accuracy** due to diverse accents or dialects. Potential delays in real-time conversion from text to sign language.



Solution

Technology Optimization:

Implement AI & ML algorithms to improve the accuracy of speech-to-sign language conversion over time. **Conduct beta testing** with diverse user groups to refine the real-time conversion system.

Challenge 3

User Adoption:

Resistance from users (teachers, students, parents) unfamiliar with digital platforms or **hesitant to adopt new technologies**.



Solution

User Training & Support:

Conduct awareness campaigns, workshops, and training sessions for **highlighting the benefits of the app**.

IMPACT AND BENEFITS



01

Large Target Audience

1.3 million people in India have hearing disabilities, with 500,000 deaf children needing specialized support.

02

ISL Resource Gap

Only 250 certified Indian Sign Language interpreters for 1.3 million deaf individuals.

03

E-Learning Growth

Indian e-learning market to grow from \$2.28B (2020) to \$10.6B (2025), at 20% CAGR.

05

Market Opportunity

Addressing ISL gaps and leveraging e-learning growth can empower students and align with government efforts.

04

Government Support

Initiatives promoting digital education and inclusivity.

Challenges faced:

1. At an international conference, experts noted that only 5% of hearing-impaired children receive basic schooling and just 1% access quality education. They stressed that daily learning challenges arise from a **lack of educational resources** in **Indian languages**. [Deafness in India – Wikipedia](#)
2. [Challenges faced by deaf and hard of hearing students in the classroom - Hear For You](#)

Existing Technology :

1. [Frontiers | Benefits and development of assistive technologies for Deaf people's communication: A systematic review \(frontiersin.org\)](#)
2. Teachers can foster an inclusive classroom for deaf and hard of hearing students by collaborating with sign language interpreters and using **assistive technologies**. [Challenges Deaf Students In The Classroom Face | Nagish](#)